PROJECT AND TEAM INFORMATION

Project Title

GRAPH BASED FRIEND AND OPPORTUNITY RECOMMENDATION SYSTEM

Team Name: *Team #*	*Tech-mates* *DSCPP-III-2026-T209*
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PHASE-II PROJECT PROGRESS DESCRIPTION

Project Abstract

The main goal of the Graph-Based Friend and Opportunity Recommendation System is to help students find relevant friend,connections,and useful academic or professional opportunities based on their interests, skills,courses, projects, workshops,seminar and competitions.The system is developed using C++ and applies Object-Oriented Programming (OOP) concepts to organize and manage user and opportunity data efficiently.C++ concepts such as classes, objects, constructors, functions, encapsulation and inheritance can be used to structure the different components of system and make the program modulr and easy to manage.

The project demonstrates the practical application of Data Structures and Algorithms in solving a real-world recommendation problem.A graph is used to represent user and their relationships,while BFS and DFS help explore connections and identify potential friends.Hashing is used for efficient searching and retrival of information,and a priority queue/max-heap helps rank recommendations according to their relevance.

Overall,the system aims to provide personalized recommendations that help students connect with suitable people,discover useful opportunities,improve their skills, and participate in relevant academic,professional activities while demonstrating the practical use of C++, OOP, and DSA concepts.

Updated Project Approach and Architecture

Our project,the Graph Based Friend and Opportunity Recommendation System,is being developed using C++ and Object-Oriented Programming (OOP) concepts.The current approach focuses on creating a modular system that recommends potential friend and relevant academic or professional opportunities based on users skills, interests,courses,projects and experiences.

1. **System Design**: The system is divided into modules for user profile management,graph construction, friend recommendation, opportunity matching and result display.Users and their relationships are represented using a graph,where vertices represent users and edges represent connections.

2. **Algorithms and Data Structures**: BFS and DFS are used to explore user connection and identify potential friends.Hashing is used for efficient searching and retrieval of user information, while a priority queue/max-heap helps rank recommendations according to their relevance.Vectors and other suitable data structures are used to store and manages information.

3. **Programming Language and Libraries**:The project uses C++ as the primary programming language.OOP concepts such as classes, objects, encapsulation, and functions help organize the code.Standard C++ libraries, including <iostream>, <vector>, <queue>, <unordered map>, and <string>,support input/output, data storage, graph traversal, hashing and text handling.

4. **Communication Protocols**: The current implementation is designed as a standalone C++ application, so network communication protocols , such as HTTP and TCP/IP are not required for its core functionality.

5. **Testing and Validation**: The system is tested using sample user profiles and opportunity data to verify graph traversal,matching,searching and recommendation ranking.

This approach aims to develop a functional and efficient prototype that demonstrates the practical application of C++, OOP and DSA in a real-world recommendation system.

Tasks Completed

Task Completed	Team Member
Module 1 - User and Account Management	Aaradhya
Module 2 - Friendship/Graph Management	Akshra
Module 3 - Friend Recommendation	Vanshika
Module 4 - Opportunity Management	Akshra
Module 5 - Opportunity Recommendation	Deepak
Module 6 - Opportunity Organization	Aaradhya
Module 7 - Data Storage and Lookup	Aaradhya

Challenges/Roadblocks

During the development of the Graph-Based Friend and Opportunity Recommendation System,we have faced several challenges.One of main challenges is designing a graph structure to represent users and their connections efficiently.Another challenge is implemention of BFS and DFS correctly to explore connections and recommend suitable friends.

We also need to identify common interests,skills,courses, and projects to generate relevant recommendations. Implementing hashing for faster searching and a priority queue/max-heap for ranking recommendations requires careful planning and testing.Additionally, organizing the program using C++ and OOP concepts such as classes and objects, is important to keep the code structured and manageable. To overcome these

challenges we plan to divide the project into smaller modules and implement each feature step by step. We will test the graph traversal algorithms using sample data,verify the recommendation results and debug errors as they arise.We will also use appropriate C++ , data structures and apply OOP concepts to improve code organization.Regular testing and team collaboration will help us identify problems early and improve the overall performance and accuracy of the system.

Tasks Pending

Task Pending	Team Member (to complete the task)
-Database integration	Deepak
-Frontend	Vanshika

Project Outcome/Deliverables

The Graph Based Friend and Opportunity Recommendation System aims to provide a platform , where students can discover meaningful connections and relevant academic or professional opportunities.The project demonstrates the practical application of C++, Object-Oriented Programming (OOP) and Data Structures and Algorithms (DSA) to develop an efficient and user friendly recommendation system.
The key outcomes and deliverables of the project are:
1. **Friend Recommendation**: Identify potential friend based on common interests,skills,courses and projects.
2. **Opportunity Recommendation**: Suggest relevant workshops,seminar,competitions and learning opportunities based on users profiles.
3. **Graph-Based Representation**: Represent user and their relationships using graph to explore direct and indirect connections.
4. **Algorithm Implementation**: Use BFS and DFS for graph traversal, hashing for efficient searching and priority queues/max-heaps for ranking recommendation.
5. **C++ and OOP Implementation**: Apply classes, objects,encapsulation and functions to organize the program into manageable component.

Progress Overview

The Graph-Based Friend and Opportunity Recommendation System has made significant progress with all seven core modules completed.The project is currently focused on integrating the remaining components to develop a complete and user friendly application.

COMPLETED MODULES:
1. User and Account Management: Managing user profile and account information.
2. Friendship / Graph Management: Creating and managing user connection.
3. Friend Recommendation: Identification of potential friends based on common attributes.
4. Opportunity Management: Adding and managing learning opportunities.
5. Opportunity Recommendation: Suggesting relevant opportunities to user.
6. Opportunity Organization: Organizing opportunities for easier access and retrival.
7. Data Storage and Lookup: Managing and retriving information , using appropriate data structures.

PENDING TASKS:
- Database Integration: Connecting the system to a database for persistent data storage and retrieval.
- Frontend Development: Developing a user-friendly interface to improve accessibility and interaction.

Schedule Status: The core functionality is progressing as planned with all seven modules completed. Database integration and frontend development are the remaining major tasks.These will be prioritized next, followed by system integration,testing and debugging to ensure that all modules work together correctly.

Codebase Information

REPOSITORY LINK-
https://github.com/UPaaradhya/Graph-based-Friend-and-Opportunity-Recommendation-system.git
BRANCHES-
main, user-authentication ,friend-graph, recommendations and modularisation.
IMPORTANT COMMITS-
1. Add project source code Establishes the starting version.
2.Add user registration and login system
3.Implement graph-based friendship management
4.Add BFS DFS and mutual friend discovery
5.Implement friend recommendation scoring
6.Add skill and interest based opportunity matching
7.Add file-based data persistence
8.Refactor main.cpp into modular components

Testing and Validation Status

Test Type	Status (Pass/Fail)	Notes
1.User registration test.	Pass	1.The user account is created successfully , and assigned a unique user ID.
2.User login test.	Pass	2.The user is authenticated and directed to the dash-board.
3. Invalid login test.	Pass	3.Access is denied and an appropriate message is displayed.
4.Friend recommendation and management test.	Pass	4.Suitable users are rec-ommended according to the implemented scoring logic.The friendship is recorded correctly in the graph.
5.Opportunity recommendation test.	Pass	5.Matching opportunities are displayed according to the implemented matching logic.
6.Data persistence test.	Pass	6.Previously saved data is loaded successfully

Deliverables Progress

The project has completed its seven core modules and is currently focused on database integration and frontend development.The status of each key deliverable is summarized below.

1. **User and Account Management — Completed:** User profiles , account-related functions have been implemented.
2. **Graph-Based Friendship Management — Completed:** Graphs structures are implemented to represent and manage user connections.
3. **Friend Recommendation — Completed:** The system identifies potential friend based on shared attributes and connections.
4. **Opportunity Management — Completed:** Functions for adding and managing learning opportunites are implemented.
5. **Opportunity Recommendation — Completed:** The system recommends relevant opportunities based on users information.
6. **Opportunity Organization — Completed:** Opportunities are organized for efficient access and management.
7. **Data Storage and Lookup — Completed:** Data handling , lookup functionality using appropriate data structures has been implemented.
8. **Database Integration — Pending:** Connecting the application to a database for persistent data storage and retrival remains to be done.
9. **Frontend Development — Pending:** Developing a user friendly graphical interface is yet to begin or be completed.
10. **System Integration and Testing — Pending:** The final integrated system , will be tested and debugged after database and frontend integration.

Overall the core recommendation functionality is completed, while database connectivity, frontend development and final system testing remain as the next priorities.